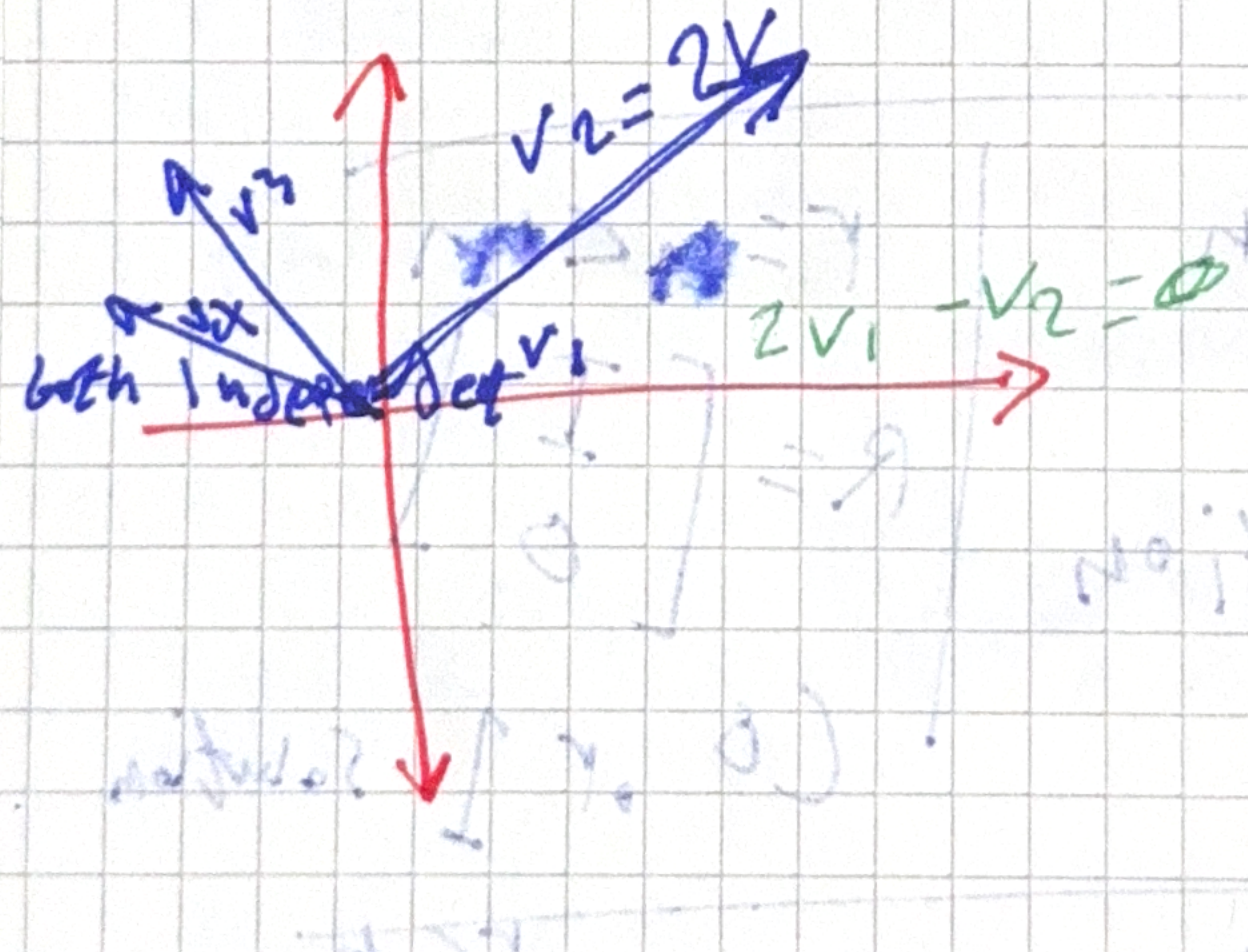
Independence Basis and dimensio

Suppose A is m by n, with $m<n$
Then there are non zero solutions to $Ax=0$
(more unknown than equations)
Reason: there will be free variables!!

Independence vectors $x_1, x_2, \dots x_n$ ~~are indepen~~
are independent if no combination gives the zero vector (except all $c_i = 0$ comb)

$$c_1x_1 + c_2x_2 \cdots + c_nx_n \neq 0$$

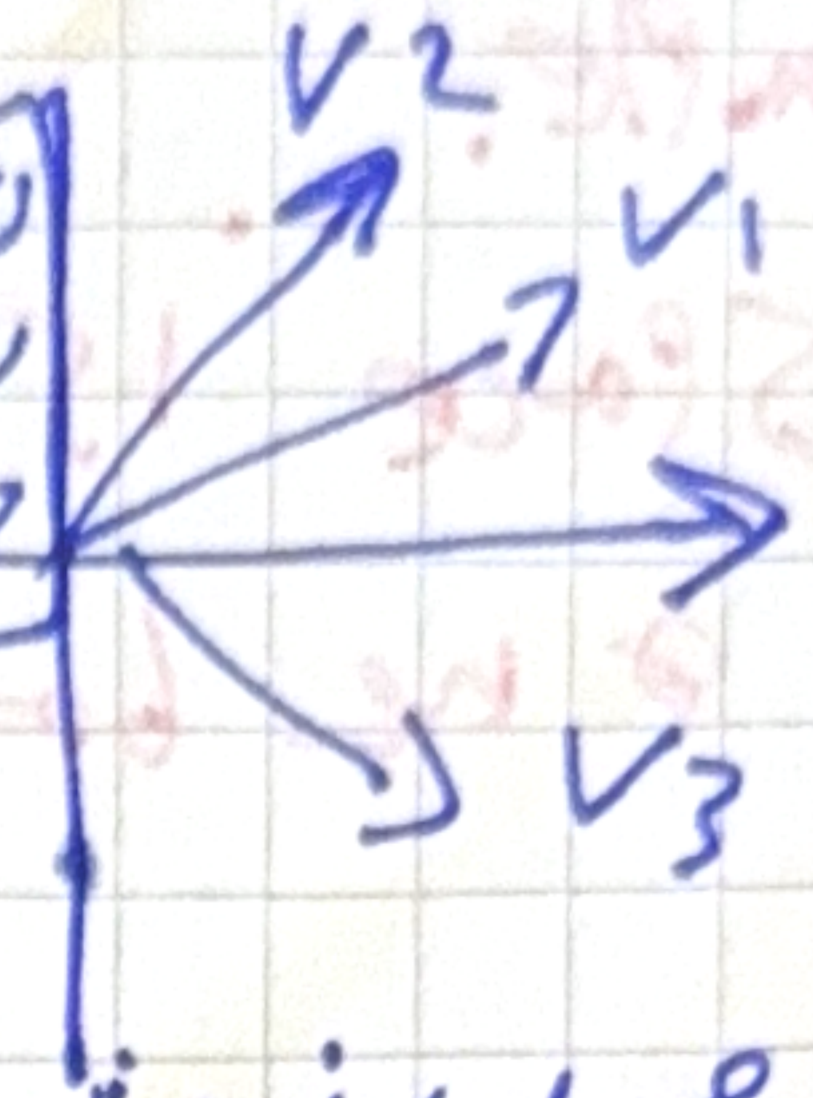


$2v_1 - v_2 = 0$

Indepandence

$$A = \begin{array}{c} \\ \\ \end{array}\begin{array}{ccc} v_1 & v_2 & v_3 \\ \end{array}$$

$$A = \left[\begin{array}{ccc} 2 & 1 & 2.5 \\ 1 & 2 & -1 \end{array}\right] \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$



* the column are dependent if it has a column in the null space

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Repeat when $v_1, \ldots, v_n$ are colums of A

They are independent if null space of A is $\{\text{zero vector}\}$

they are dependent if $Ac=0$ for some non zero c.

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rank=n $N(A)=\{0\}$

rank<n yes free variables

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vectors $v_1 \ldots v_n$ span a space means: the space consists of all combs of those vectors

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Basis for a space is a sequence of vectors $v_1, v_2, \ldots, v_d$ with 2 properties

1- they ar independent

2. they span the space

Example:

Space is R^3

one basis is $\begin{bmatrix}1\\0\\0\end{bmatrix}, \begin{bmatrix}0\\1\\0\end{bmatrix}, \begin{bmatrix}0\\0\\1\end{bmatrix}$

Another basis $\begin{bmatrix}1\\1\\2\end{bmatrix}, \begin{bmatrix}2\\2\\5\end{bmatrix}, \begin{bmatrix}3\\3\\8\end{bmatrix}$

R^n n vectors give basis if
the nxn matrix with thos cols
is

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the nxn matrix with those cols
is inventible

Given a spaze: ~~cols~~
Every basis for the space
has the same number of vetors

Def Dimenson of the space

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space C(A) N(A)

$$\begin{bmatrix}1 & 2 & 3 & 1\\1 & 1 & 2 & 1\\1 & 2 & 3 & 1\end{bmatrix} \quad \begin{bmatrix}-1\\-1\\1\\0\end{bmatrix}$$

rank(A) = # pivot columns = dimension of C(A)

another basis for C(A) $\begin{bmatrix} 2 \\ 2 \\ 7 \end{bmatrix} \left\{ \begin{bmatrix} 1 \\ 5 \\ 7 \end{bmatrix}, \begin{bmatrix} -1 \\ -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right.$

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dim C(A) = r

dim N(A) = # free variable
= n − r